THE UNIVERSITY OF QUEENSLAND
AUSTRALIA

This exam paper must not be removed from the venue

Venue ____________________
Seat Number __________
Student Number |__|__|__|__|__|__|__|__|
Family Name ____________________
First Name ____________________

School of Mathematics & Physics
Semester 1 Examinations, 2025
MATH1061 Discrete Mathematics

This paper is for St Lucia Campus students.

Examination Duration: 120 minutes

Planning Time: 10 minutes

Exam Conditions:

- No written or printed material permitted
- Casio FX82 series or UQ approved and labelled calculator only
- During Planning Time - Students are encouraged to review and plan responses to the exam questions

Materials Permitted in the Exam Venue:
(No electronic aids are permitted e.g. laptops, phones)

None

Materials to be supplied to Students:
Additional exam materials (e.g. answer booklets, rough paper) will be provided upon request.

None

Instructions to Students:
If you believe there is missing or incorrect information impacting your ability to answer any question, please state this when writing your answer.

Complete all examination questions in the space provided on this examination paper. If more space is required, use the backs of pages. The final page is a formula sheet, which you may detach. Nothing written on the formula sheet will be marked. Answers will be marked based on correctness, quality of expression, depth of understanding demonstrated, and clarity of explanation.

For Examiner Use Only

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	
13	
14	

Total __________
out of 65

1. Let p, q and r be statements. Use the laws of logical equivalence to show that

$$p \to (q \vee r) \equiv (p \wedge \sim q) \to r.$$

(3 marks)

2. For each of the following statements, write down the negation of the statement, and state which of the original or the negation is true. Justify your answers.

(a) $\exists x \in \mathbb{Z}^+$ such that $\forall y \in \mathbb{Z}^+$, $x \geq y$.

(2 marks)

(b) $\forall r, s \in \mathbb{R}$, if $rs \in \mathbb{Q}$, then $r \in \mathbb{Q}$ and $s \in \mathbb{Q}$.

(2 marks)

3. For each of the following statements, either prove the statement or give a counterexample to disprove the statement.

(5 marks)

(a) For all $a, b, c \in \mathbb{Z}^+$, if $a \mid b$ and $b \mid c$, then $a \mid (b + c)$.

(b) For all $a, b, c \in \mathbb{Z}^+$, if $a \mid (b + c)$, then $a \mid b$ or $a \mid c$.

4. Use mathematical induction to prove that $3^n + 5 < 4^n$ for each integer $n \geq 2$.

(5 marks)

5. (a) Use the Euclidean algorithm to determine $\gcd(931, 301)$. Show your working.

(2 marks)

(b) Write down the unique prime factorisation of 931 in standard form.

(1 mark)

(c) How many positive divisors does 931 have? (Hint: use part (b)).

(1 mark)

6. Recall that $\emptyset$ denotes the empty set. Let $A = \{1, 2\}$, $B = \{-1, 2\}$, and $C = \{\emptyset, \{-1\}\}$.

Write down the following sets, by listing their elements and using appropriate brackets.

(5 marks)

(a) $A \cup B$

(b) $B \cap C$

(c) $\mathcal{P}(C)$

(d) $(A \times B) - (B \times B)$

(e) $\{X \mid (X \subseteq A) \wedge (X \in C)\}$

7. (a) Consider the function $f : \mathbb{Z}^+ \times \mathbb{Z}^+ \to \mathbb{Z}^+$ where for each element $(m, n) \in \mathbb{Z}^+ \times \mathbb{Z}^+$ we define

$$f((m, n)) = 2^m 3^n.$$

(i) Determine $f((3, 1))$.

(1 mark)

(ii) Determine the pre-image of $\{36\}$.

(2 marks)

(iii) Prove that the function f is one-to-one.

(4 marks)

(b) Let A and B be sets. Which one of the following statements is implied by the existence of a one-to-one function $g : A \to B$? (circle your answer)

(1 mark)

$|A| \geq |B|$ $\qquad$ $|A| \leq |B|$ $\qquad$ $|A| = |B|$ $\qquad$ $|A| \neq |B|$

8. Let ρ be the relation defined on $\mathbb{Z}^+$ as follows. For all $a, b \in \mathbb{Z}^+$, $a\rho b$ if and only if

$$a^2 - b^2 \equiv 0 \pmod 5.$$

(a) Prove that ρ is an equivalence relation.

(5 marks)

(b) How many equivalence classes does the relation ρ have?

(1 mark)

(c) List the elements of the equivalence class [1] that are greater than 1 and less than 10.

(1 mark)

9. The Cayley table for a group $(G, *)$ is given below. (You do not need to prove that this is a group.)

$*$	a	b	c	d	e	f
a	a	b	c	d	e	f
b	b	a	f	e	d	c
c	c	e	a	f	b	d
d	d	f	e	a	c	b
e	e	c	d	b	f	a
f	f	d	b	c	a	e

(a) Is this group abelian? Justify your answer.

(1 mark)

(b) Which element is the identity of this group? Justify your answer.

(1 mark)

(c) Which element is the inverse of the element b? Justify your answer.

(1 mark)

(d) Write down the elements of the subgroup $< e >$.

(1 mark)

10. Solve the equation $9x - 7 = 4$ in the field $(\mathbb{Z}_{23}, +, \cdot)$, and hence determine the smallest non-negative integer y such that $9y - 7 \equiv 4 \pmod{23}$. *Hint:* $9 \cdot 18 \equiv 1 \pmod{23}$.

(3 marks)

11. Seven friends, named Alice, Bob, Charlie, Delia, Edward, Fred and George, are meeting for dinner and each person is bringing one dish to share.

(a) For the dinner, two people will bring desserts and the other five will bring savoury dishes. How many different choices are there for the two people who will bring desserts?

(1 mark)

(b) After dinner the seven friends stand in a line for a photograph. How many different arrangements are there of the seven friends standing in a line?

(1 mark)

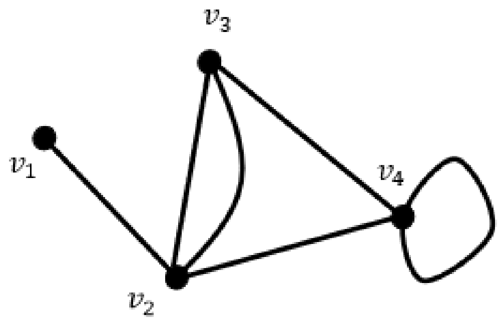
(c) How many different arrangements are there of the seven friends standing in a line such that Alice is not next to Charlie and Bob is not next to Delia.

(4 marks)

12. Let $S = \{x \in \mathbb{Z} \mid 1 \leq x \leq 99\}$, so S is the set of integers from 1 to 99. How many integers must you pick from S in order to be sure that at least one of them is divisible by 3? Justify your answer.

(3 marks)

13. Let G be the graph shown below.



(a) Write down an adjacency matrix for G. Make sure you clearly label the rows and columns of your matrix.

(2 marks)

(b) Does G have an Euler circuit? If yes, write it out as a sequence of vertices. If not, determine the fewest number of edges that would need to be added to G to create a new graph that does have an Euler circuit.

(2 marks)

14. A tree has 11 vertices, of which exactly four vertices have degree 1 and exactly one vertex has degree 4. Determine the degrees of the remaining six vertices. Show your working.

(4 marks)

END OF EXAMINATION

MATH1061/MATH7861 Final Examination Formula Page

Logical Equivalences

Given any statement variables p, q and r, a tautology $\mathbf{t}$ and a contradiction $\mathbf{c}$, the following logical equivalences hold.

Commutative laws	$p \wedge q \equiv q \wedge p$	$p \vee q \equiv q \vee p$
Associative laws	$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	$(p \vee q) \vee r \equiv p \vee (q \vee r)$
Distributive laws	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
Identity laws	$p \wedge \mathbf{t} \equiv p$	$p \vee \mathbf{c} \equiv p$
Negation laws	$p \vee \sim p \equiv \mathbf{t}$	$p \wedge \sim p \equiv \mathbf{c}$
Double negative laws	$\sim (\sim p) \equiv p$	
Idempotent laws	$p \wedge p \equiv p$	$p \vee p \equiv p$
Universal bound laws	$p \vee \mathbf{t} \equiv \mathbf{t}$	$p \wedge \mathbf{c} \equiv \mathbf{c}$
De Morgan's laws	$\sim (p \wedge q) \equiv \sim p \vee \sim q$	$\sim (p \vee q) \equiv \sim p \wedge \sim q$
Absorption laws	$p \vee (p \wedge q) \equiv p$	$p \wedge (p \vee q) \equiv p$
Negations of $\mathbf{t}$ *and* $\mathbf{c}$	$\sim \mathbf{t} \equiv \mathbf{c}$	$\sim \mathbf{c} \equiv \mathbf{t}$

Valid Argument Forms

Modus Ponens	$p \to q$ p $\therefore q$		Elimination	(a) $p \vee q$ $\sim q$ $\therefore p$	(b) $p \vee q$ $\sim p$ $\therefore q$
Modus Tollens	$p \to q$ $\sim q$ $\therefore \sim p$		Transitivity	$p \to q$ $q \to r$ $\therefore p \to r$	
Generalization	(a) p $\therefore p \vee q$	(b) q $\therefore p \vee q$	Proof by division into cases	$p \vee q$ $p \to r$ $q \to r$ $\therefore r$	
Specialization	(a) $p \wedge q$ $\therefore p$	(b) $p \wedge q$ $\therefore q$			
Conjunction	p q $\therefore p \wedge q$		Contradiction Rule	$\sim p \to \mathbf{c}$ $\therefore p$	

Set Identities For all sets A, B and C that are subsets of a universal set U:

1. Commutative Laws (a) $A \cup B = B \cup A$ (b) $A \cap B = B \cap A$
2. Associative Laws (a) $(A \cup B) \cup C = A \cup (B \cup C)$ (b) $(A \cap B) \cap C = A \cap (B \cap C)$
3. Distributive Laws (a) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ (b) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
4. Identity Laws (a) $A \cup \emptyset = A$ (b) $A \cap U = A$
5. Complement Laws (a) $A \cup A^c = U$ (b) $A \cap A^c = \emptyset$
6. Double Complement Law $(A^c)^c = A$
7. Idempotent Laws (a) $A \cup A = A$ (b) $A \cap A = A$
8. Universal Bound Laws (a) $A \cup U = U$ (b) $A \cap \emptyset = \emptyset$
9. De Morgan's Laws (a) $(A \cup B)^c = A^c \cap B^c$ (b) $(A \cap B)^c = A^c \cup B^c$
10. Absorption Laws (a) $A \cup (A \cap B) = A$ (b) $A \cap (A \cup B) = A$
11. Complement of U and $\emptyset$ (a) $U^c = \emptyset$ (b) $\emptyset^c = U$
12. Set Difference Law $A - B = A \cap B^c$

The Quotient-Remainder Theorem Given any integer n and any positive integer d, there exist unique integers q and r such that $n = dq + r$ and $0 \leq r < d$.

Schröder-Bernstein Theorem For all sets A and B, if $|A| \leq |B|$ and $|A| \geq |B|$ then $|A| = |B|$.

Binomial Theorem Given any real numbers a and b, and any nonnegative integer n,

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k.$$